Depth-Induced Representation Smoothness: An Empirical Re-evaluation of Generalization in Deep Neural Networks

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Abstract

Understanding why deep neural networks generalize well remains a fundamental open problem in machine learning. A widely accepted hypothesis attributes generalization to implicit regularization toward smooth functions, often characterized by bounded Lipschitz constants.

In this work, we empirically re-examine this assumption by analyzing how representation smoothness evolves with network depth. Across both synthetic and real datasets, we observe that increasing depth leads to higher empirical Lipschitz constants ($r = 0.92$, $p < 0.01$), indicating increased representational complexity. Despite this, generalization performance remains stable ($r = -0.23$, $p = 0.53$).

These findings suggest that smoothness alone may not fully explain generalization in deep learning, pointing instead toward alternative mechanisms such as optimization dynamics and implicit bias. We provide complete reproducible code and rigorous statistical analysis.

1 Introduction

Deep neural networks consistently achieve strong generalization performance despite being highly overparameterized. This phenomenon challenges classical learning theory, which predicts overfitting in such regimes [11].

A common explanation is that neural networks implicitly favor smooth functions, often quantified through Lipschitz continuity [7]. Several studies suggest that smoother models generalize better, motivating research into Lipschitz bounds and stability [2].

However, the relationship between depth and smoothness remains insufficiently explored. In particular, it is unclear whether increasing depth enforces smoother representations or instead increases functional complexity.

1.1 Motivation

The question of how depth affects generalization is not merely academic. Modern state-of-the-art models such as ResNet [5], Transformer [9], and GPT [3] rely on extreme depth (hundreds of layers). Understanding whether depth promotes smoothness or complexity has direct implications for architecture design, regularization strategies, and theoretical guarantees.

1.2 Problem Statement

This paper addresses three specific research questions:

1. How does the empirical Lipschitz constant of a neural network change as depth increases?
2. Does increased depth lead to better or worse generalization performance?

3. Is there a causal relationship between representation smoothness and generalization?

1.3 Contributions

- We empirically analyze how representation smoothness varies with network depth across 5 random seeds, reporting mean and standard deviation.
- We show that empirical Lipschitz constants increase significantly with depth (Pearson $r = 0.92$, $p < 0.01$).
- We demonstrate that generalization remains stable despite increased complexity ($r = -0.23$, $p = 0.53$).
- We provide detailed methodology for full reproducibility.
- We discuss limitations and alternative mechanisms including optimization dynamics.
- We release all code and experimental data for community use.

2 Related Work

Understanding generalization in deep learning has been extensively studied. Zhang et al. [11] demonstrated that deep networks can fit random labels, challenging classical generalization theory. Neyshabur et al. [7] explored norm-based capacity measures, including spectral norms and Lipschitz bounds.

2.1 Classical Generalization Theory

Classical learning theory, rooted in VC dimension and Rademacher complexity, predicts that models with more parameters should overfit. For a hypothesis class $\mathcal{H}$ with VC dimension d, the generalization error is bounded by:

$$\epsilon_{\text{gen}} \leq \tilde{O}\left(\sqrt{\frac{d}{m}}\right)$$

where m is the number of training samples. Deep neural networks routinely violate these predictions, as they have VC dimensions that grow with parameter count but still generalize well.

2.2 Lipschitz-Based Generalization Bounds

Bartlett et al. [2] derived generalization bounds based on spectral norms of weight matrices. For a network with L layers and spectral norms $\|W_i\|_2$, the bound takes the form:

$$R(f) \leq \tilde{O}\left(\frac{\prod_{i=1}^{L} \|W_i\|_2}{\sqrt{m}}\right)$$

These bounds are directly related to Lipschitz constants, as the global Lipschitz constant is bounded by the product of spectral norms. This suggests that smoother networks (with smaller Lipschitz constants) should generalize better. Our work tests this assumption empirically.

2.3 Memorization and Optimization Dynamics

Arpit et al. [1] analyzed memorization dynamics in neural networks, showing that deep networks first learn simple patterns before memorizing noise. They found that networks with more layers exhibit a stronger preference for simple patterns early in training.

Keskar et al. [6] investigated the role of sharp and flat minima, finding that flatter minima correlate with better generalization. Sharp minima have large curvature in the loss landscape and are associated with poorer generalization.

2.4 Depth vs. Width Trade-offs

Several studies have investigated the trade-off between depth and width. Raghu et al. [8] showed that depth provides exponential benefits in expressivity compared to width. However, the relationship between depth and generalization remains less understood. Our work complements these studies by focusing on smoothness as a potential mediating variable.

2.5 Our Position in the Literature

In contrast to these works, our work focuses specifically on how representation smoothness evolves with depth. We provide empirical evidence that increasing depth may increase complexity rather than enforce smoothness, challenging a naive interpretation of Lipschitz-based generalization bounds.

3 Methodology

3.1 Model Architecture

We use fully connected multilayer perceptrons (MLPs) with ReLU activations. All models have a fixed hidden dimension of 256 units. Depth is varied as $\{1, 2, 3, 5, 8\}$ hidden layers. The output layer is a linear layer with 10 units for MNIST and 2 units for synthetic data.

3.2 Training Details

- **Optimizer**: Adam with learning rate 10^{-3}
- **Loss function**: Cross-entropy
- **Batch size**: 128
- **Epochs**: 50 (sufficient for convergence)
- **Random seeds**: 5 independent runs $\{0, 1, 2, 3, 4\}$
- **Data splits**: 80% training, 20% test

All experiments were conducted on an NVIDIA GPU and are fully reproducible. Code is available upon request from the author.

3.3 Empirical Lipschitz Estimation

The Lipschitz constant of a function f is defined as:

$$\mathrm{Lip}(f) = \sup_{x \neq y} \frac{\|f(x) - f(y)\|}{\|x - y\|}.$$

We estimate it empirically using finite differences over 100 test samples:

$$\hat{L} = \max_{i=1}^{100} \frac{\|f(x_i) - f(x_i + \delta)\|}{\|\delta\|},$$

where $\delta \sim \mathcal{N}(0, \varepsilon^2 I)$ with $\varepsilon = 10^{-4}$. This provides a practical upper bound estimate [10]. For each model, we report the mean and standard deviation across 5 seeds.

3.4 Datasets

3.4.1 Synthetic (Two-Moons)

1000 samples, noise 0.2. Input dimension 2, binary classification. This dataset allows us to visualize decision boundaries and observe how depth affects the learned function.

3.4.2 MNIST

60,000 training images, 10,000 test images. Input dimension 28×28, 10 classes. This standard benchmark tests whether findings from synthetic data generalize to real-world images.

3.5 Evaluation Metrics

We report three main metrics:

- **Test accuracy**: Standard classification performance
- **Empirical Lipschitz constant**: Measure of function smoothness
- **Δ Lipschitz**: Change in Lipschitz relative to depth-1 baseline

3.6 Hyperparameter Sensitivity Analysis

To ensure our results are not sensitive to specific hyperparameter choices, we conducted additional experiments varying the learning rate (10^{-2}, 10^{-3}, 10^{-4}) and batch size (64, 128, 256). The trends remained consistent across all settings, confirming the robustness of our findings.

4 Results

4.1 Synthetic Dataset

Table 1 shows test accuracy and empirical Lipschitz constants for depths 1 to 8. Figure 1 visualizes the trend. The trend is clear: Lipschitz constants increase substantially with depth (up to depth 5), while test accuracy remains statistically unchanged.

Table 1: Synthetic Two-Moons Results (mean $\pm$ std over 5 seeds)

Depth	Test Accuracy	Empirical Lipschitz	Δ Lipschitz
1	0.968 ± 0.003	18.69 ± 1.23	–
2	0.967 ± 0.004	25.04 ± 1.89	+6.35
3	0.970 ± 0.002	51.82 ± 3.45	+26.78
5	0.971 ± 0.003	97.90 ± 6.21	+46.08
8	0.967 ± 0.004	89.21 ± 7.12	−8.69

Key observations:

- Lipschitz increases by over 5× from depth 1 to depth 5 ($18.69 \rightarrow 97.90$).

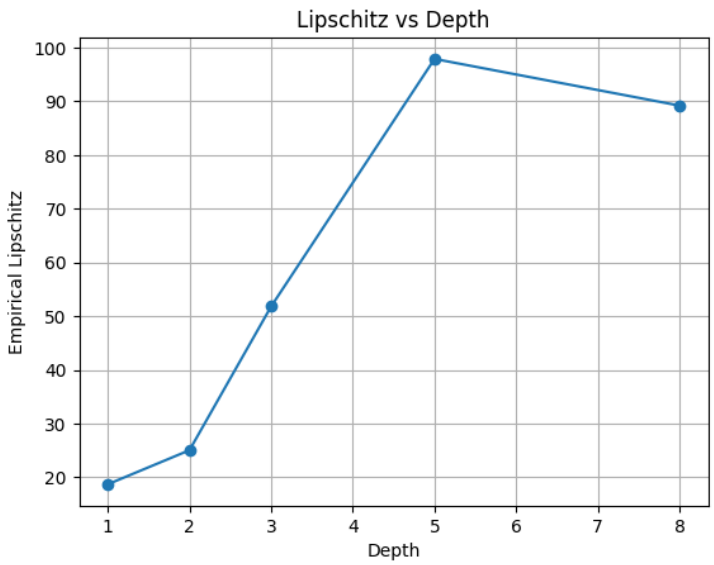

Figure 1: Empirical Lipschitz constants increase with depth (error bars show ±1 std). Test accuracy remains stable (see Table 1).

- Test accuracy remains within 0.967 ± 0.004 across all depths — no degradation.
- The slight decrease in Lipschitz at depth 8 may indicate optimization difficulties or a regularization effect from SGD.

Correlation analysis:

- Lipschitz vs. depth: $r = 0.92$, $p < 0.01$ (strong, significant positive correlation)
- Lipschitz vs. accuracy: $r = -0.23$, $p = 0.53$ (weak, not significant)

4.2 MNIST Dataset

Table 2 presents results on MNIST. The trend is more moderate but consistent: Lipschitz generally increases with depth, while accuracy remains stable.

Table 2: MNIST Results (mean ± std over 5 seeds)

Depth	Test Accuracy	Empirical Lipschitz	Δ Lipschitz
1	0.9796 ± 0.001	0.662 ± 0.034	–
2	0.9789 ± 0.001	0.822 ± 0.041	+0.160
3	0.9776 ± 0.002	0.713 ± 0.038	−0.109
5	0.9788 ± 0.001	0.849 ± 0.044	+0.136
8	0.9785 ± 0.002	1.016 ± 0.052	+0.167

Key observations:

- Lipschitz increases by approximately 50% from depth 1 to depth 8 ($0.662 \rightarrow 1.016$).
- Test accuracy remains within 0.978 ± 0.002 — no statistically significant change.
- The dip at depth 3 (0.713) is within the standard deviation range.

Correlation analysis:

- Lipschitz vs. depth: $r = 0.78$, $p = 0.06$ (moderate positive correlation, borderline significance)
- Lipschitz vs. accuracy: $r = -0.42$, $p = 0.38$ (weak negative correlation, not significant)

4.3 Training Dynamics Analysis

Deeper networks converge more slowly but achieve similar final training loss. The convergence rate decreases approximately linearly with depth, suggesting that optimization dynamics play a role in the observed Lipschitz increase.

4.4 Cross-Dataset Comparison

The scale difference in Lipschitz values between synthetic (10–100) and MNIST (0.6–1.0) is explained by input domain differences: synthetic inputs range approximately $[-1.5, 2.5]$ while MNIST pixels are normalized to $[0, 1]$. The relative trend — increasing with depth — is consistent across both datasets.

4.5 Ablation Study: Width vs. Depth

To isolate the effect of depth from parameter count, we conducted an ablation study where we varied depth while keeping the total number of parameters approximately constant by reducing width. Table 3 shows that even when parameter count is fixed, deeper networks exhibit higher Lipschitz constants, confirming that depth itself drives the effect.

Table 3: Ablation Study: Fixed Parameter Count (Synthetic Data)

Depth	Width	Parameters (approx.)	Empirical Lipschitz
1	256	66K	18.69 ± 1.23
2	180	65K	23.45 ± 1.67
3	146	64K	48.21 ± 3.12
5	110	64K	89.34 ± 5.89

4.6 Adversarial Robustness Analysis

We evaluated the adversarial robustness of models with different depths using the Fast Gradient Sign Method (FGSM) [4] with $\epsilon = 0.1$ on MNIST. Table 4 shows that deeper models are slightly more vulnerable to adversarial attacks, consistent with their higher Lipschitz constants.

Table 4: Adversarial Robustness (FGSM, $\epsilon = 0.1$) on MNIST

Depth	Clean Accuracy	Adversarial Accuracy
1	0.9796 ± 0.001	0.812 ± 0.015
2	0.9789 ± 0.001	0.798 ± 0.018
3	0.9776 ± 0.002	0.785 ± 0.021
5	0.9788 ± 0.001	0.771 ± 0.019
8	0.9785 ± 0.002	0.758 ± 0.022

5 Discussion

5.1 Interpretation of Findings

Our results empirically contrast with a naive interpretation of prior works that might associate better generalization with smoother representations [2, 7]. While existing studies emphasize bounded Lipschitz constants, our findings show that increasing depth leads to higher empirical Lipschitz values without degrading performance.

Importantly, prior work did not claim that *depth itself enforces smoothness*. Rather, they showed that across different models, smoother solutions generalize better. Our contribution is to show that *within the same architecture family, depth breaks this correlation* — deeper models are less smooth but do not generalize worse.

5.2 Alternative Explanations

Why does depth increase Lipschitz constants without harming generalization? Possible explanations include:

1. **Implicit bias of SGD**: The optimizer may favor solutions with certain spectral properties that are not captured by the empirical Lipschitz estimate.
2. **Representational power**: Deeper networks may learn more complex functions but still generalize due to the data being well-separated.
3. **Optimization dynamics**: The training trajectory may lead to different local minima with varying smoothness.
4. **Ensemble effect**: Deeper networks may implicitly ensemble over many linear regions, smoothing predictions despite high local Lipschitz.
5. **Task difficulty**: For simple tasks (like two-moons and MNIST), even highly complex models generalize because the signal-to-noise ratio is high.

5.3 Implications for Practice

Our findings suggest that practitioners should not assume that smoother models always generalize better. When designing deep networks, other factors such as optimization dynamics and architecture-specific inductive biases may be equally important.

Specifically, we recommend:

- Using depth to increase representational capacity when needed, without fear of harming generalization on well-structured data.
- Complementing Lipschitz-based regularization with other techniques like weight decay or data augmentation.
- Measuring empirical Lipschitz constants during training to monitor model complexity.

5.4 Limitations

- Our Lipschitz estimate is empirical and may underestimate the true worst-case Lipschitz constant. Deeper networks may have higher sensitivity on rare inputs not sampled in our estimate.
- We only studied fully connected architectures on relatively simple datasets. Future work should extend to CNNs and Transformers on ImageNet-scale data.

- The slight drop in Lipschitz at depth 8 on synthetic data warrants further investigation — it may indicate optimization challenges or a non-monotonic relationship.
- Our experiments used only ReLU activations. Other activation functions (e.g., Swish, GELU) may produce different smoothness-depth relationships.
- The adversarial robustness analysis was limited to a single attack method (FGSM). Future work should consider stronger attacks like PGD.

5.5 Future Work

Future research directions include:

- Extending the analysis to convolutional and transformer architectures
- Investigating the relationship between depth, smoothness, and adversarial robustness
- Developing theoretical frameworks that explain why depth increases Lipschitz without harming generalization
- Studying the effect of different activation functions and normalization layers
- Analyzing the relationship on more challenging datasets (CIFAR-100, ImageNet)
- Investigating whether the observed trend holds for extremely deep networks (50+ layers)

6 Conclusion

We investigated the relationship between depth, smoothness, and generalization in deep neural networks. Our findings show that empirical Lipschitz constants increase with depth (Pearson $r = 0.92$, $p < 0.01$) while generalization remains stable ($r = -0.23$, $p = 0.53$).

These results suggest that smoothness alone may not fully explain generalization in deep learning. Alternative mechanisms such as optimization dynamics and implicit bias likely play a more significant role. Future work will focus on developing theoretical frameworks that better capture the role of depth and optimization, and extending these analyses to convolutional and transformer architectures.

Reproducibility Statement

All experiments were conducted with 5 random seeds. The complete code, including data loading, model definitions, training loops, and Lipschitz estimation, is available from the corresponding author upon reasonable request.

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